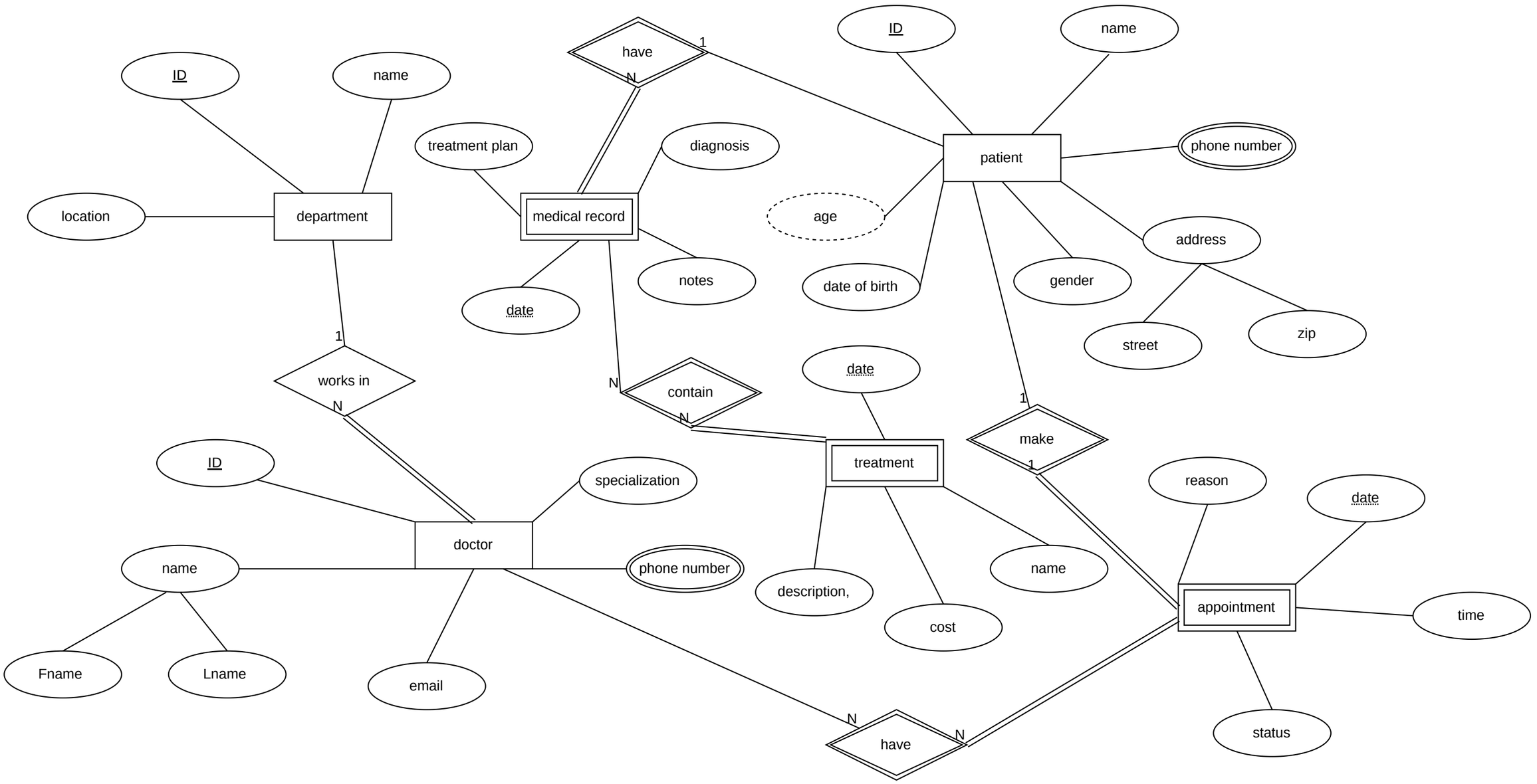
Hospital Management System

Entity Relationship Diagram (ERD) | Chen notation

Figure 1



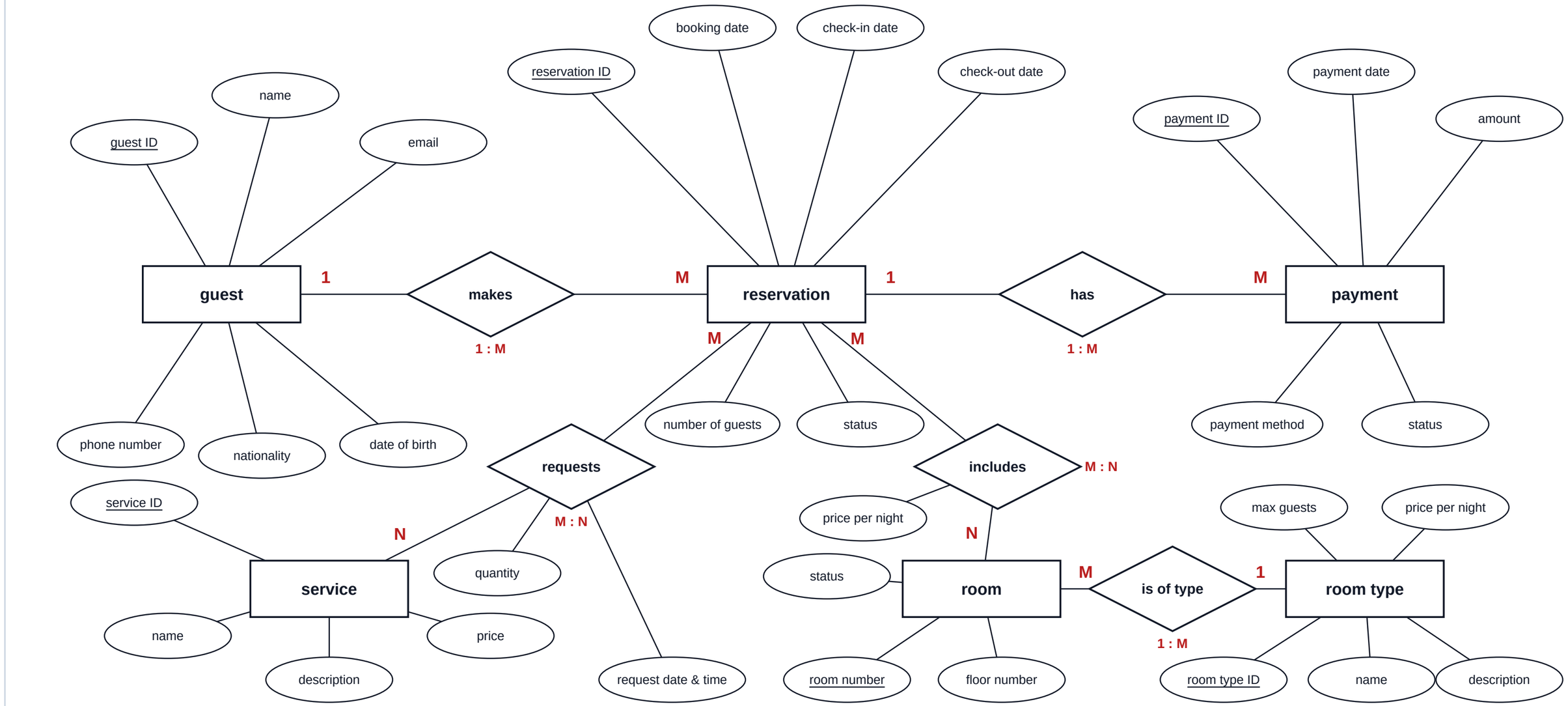
NOTATION KEY

Entity | Weak entity | Relationship | Identifying relationship | Attribute | Key attribute | Multivalued attribute | Derived attribute

Hotel Reservation & Management System

Entity Relationship Diagram (ERD) | Chen notation

Figure 2



Relationship Summary

#	Relationship	Entities	Cardinality	Type	Meaning
1	**makes**	guest - reservation	**1 : M**	**One-to-Many (1:M)**	A guest can make many reservations; each reservation belongs to exactly one guest.
2	**includes**	reservation - room	**M : N**	**Many-to-Many (M:N)**	A reservation can include one or more rooms, and a room can appear in many reservations over time. Attribute: price per night (agreed for that reservation).
3	**is of type**	room type - room	**1 : M**	**One-to-Many (1:M)**	One room type can be assigned to many rooms; every room belongs to exactly one room type.
4	**requests**	reservation - service	**M : N**	**Many-to-Many (M:N)**	A reservation can have many service requests and a service can be requested in many reservations. Attributes: quantity, request date & time.
5	**has**	reservation - payment	**1 : M**	**One-to-Many (1:M)**	A reservation can have one or more payments; each payment belongs to one reservation.